



Artificial intelligence and machine learning in finance: A bibliometric review

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ABSTRACT

This study reviewed the artificial intelligence (AI) and machine learning (ML) literature in the finance field. Using a bibliometric approach, we collected 348 articles published in 2011–2021 from journals indexed in the Scopus database. Multiple software (RStudio, VOSviewer, and Excel) were employed to analyze the data and depict the most active scientific actors in terms of countries, institutions, sources, documents, and authors. Our review revealed an upward trajectory in the publication trend starting from 2015 and found the application of AI and ML in bankruptcy prediction, stock price prediction, portfolio management, oil price prediction, anti-money laundering, behavioral finance, big data analytics, and blockchain. Moreover, the United States, China, and the United Kingdom were the top three contributors to the literature. Our results provide practical guidance to market participants, especially, fintech and finance companies, on how AI and ML can be used in their decision-making.

1. Introduction

Artificial intelligence (AI), a key technology in the 2010s, has been increasingly becoming a dominant technology in the 2020 s (Hilpisch, 2020). For example, numerous businesses have applied AI to improve their operations (Bughin et al., 2018), while many multi-national corporations (e.g., Facebook, Google, IBM, and Microsoft) have significantly invested in AI to improve their efficiency (Mhlanga, 2020). Similarly, AI applications have become an integral part of the financial services industry, especially in automating operations (Königstorfer and Thalmann, 2020a, 2020b). In particular, the use of AI applications in financial technologies (fintech) has transformed the financial services industry as a whole (Mamoshina et al., 2018).

AI applications are critical for capturing the linear and nonlinear behaviors of finance variables. Specifically, such applications allow us to solve highly nonlinear problems that traditional models cannot address. This has led to widespread applications of machine learning (ML) and deep learning techniques, a subset of AI in the finance field. Meanwhile, the literature on the application of ML and deep learning has been rapidly growing (Bahrammirzaee, 2010; Goodell et al., 2021; Gupta et al., 2020; Wong and Selvi, 1998). Thus, the present study reviews the literature on how AI applications play key a role in a number of finance areas, namely, bankruptcy prediction, stock price prediction, portfolio management, oil price prediction, anti-money laundering, behavioral finance, and big data

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analytics. Goodell et al. (2021, p. 1) found three overarching clusters of AI and ML research in finance: “1) portfolio construction, valuation, and investor behavior; 2) financial fraud and distress; and 3) sentiment inference, forecasting, and planning.” Bahrammirzaee (2010) reviewed the literature on the application of AI until 2009 and found three main areas: 1) credit evaluation; 2) portfolio management; and 3) financial prediction and planning. Our study both complements and extends Goodell et al.’s (2021) and Bahrammirzaee’s (2010) reviews by including studies up to 2021. Additionally, our review only focuses on the Scopus Q1 and Q2 finance journals.

Using a bibliometric approach, we collected 348 articles published in 2011–2021 from journals indexed in the Scopus database. Several bibliometric tests were conducted, including citation, co-citation, co-authorship, trend analysis, and keyword occurrences. Our review also focused on six research streams from the literature: 1) bankruptcy prediction and credit-risk assessment; 2) stock price prediction, portfolio management, volatility, and liquidity; 3) price prediction of oil, gold, and agriculture products; 4) anti-money laundering, anti-fraud detection, and risk management; 5) behavioral finance; and 6) big data analytics. The notable studies on bankruptcy prediction include the following: Hernandez Tinoco and Wilson (2013), who applied artificial neural networks to predict corporate bankruptcy; and Lahmri and Bekiros (2019), who used nonlinear classifiers based on big data analytics and ML approaches. Other notable studies include: Ghosh et al. (2021) and Liu et al. (2021) on stock price prediction; Verma (2021), Malliaris and Malliaris (2013), and Manogna and Mishra (2021) on price prediction of oil, gold, and agriculture products; Garcia-Bedoya et al. (2020), Jullum et al. (2020), Eling et al. (2021), and Goel and Uzuner (2016) on anti-money laundering, fraud detection, and risk management; Bhatia et al. (2020), Königstorfer and Thalmann (2020a, 2020b), and Félix et al. (2020) on behavior finance; and Hanafy and Ming (2021) and Huang et al. (2017) on big data analytics and data mining.

The remainder of this study is as follows. Section 2 discusses the data and methodology, while Section 3 analyzes the results of this study. Section 4 presents the conclusion.

2. Data and methodology

2.1. Data collection

2.1.1. The source of data collection

The Scopus database was our source for data collection, as seen in previous studies (Alshater et al., 2020; Baker et al., 2020; Goodell et al., 2021). We used this database for a number of reasons. First, it is one of the largest multidisciplinary databases of peer-reviewed academic literature. Second, it is among the top accessible databases listing the most reputable journals in the finance field. Third, it provides users with advanced search capabilities and various features for bibliometric analysis such as exporting bibliographical data based on users’ needs. Finally, for the purpose of this study, it includes a significant number of articles, compared to other options such as the Web of Science (WOS-SSCI).

2.1.2. Keyword selection strategy and refinement process

The keyword selection process was carried in two main phases in order to ensure accuracy. In phase one, we examined the AI and ML online dictionaries and glossaries to select the most relevant keywords. In phase two, we employed a Scopus search for the terminologies of ML and AI in two steps. The first step was explorative. We also utilized the Scopus keyword feature to select further keywords in our final query up to the permitted number of characters on the Scopus search bar. In the second step, we manually checked each keyword.

After preparing the query, the search was conducted on both the TITLE and the KEYWORD criteria, as shown in Table 1. In order to obtain the most relevant articles and reviews for the scope of our study, we limited the results to articles and reviews in English journals in the finance field, which yielded 5,699 documents. Although our preliminary dataset contained a large amount of data, we conducted several online refinements. First, we limited the time frame of the articles and reviews to 2011–2021. This was based on the fact that until 2011, the number of AI and ML studies in the finance field was in the single digits (see Goodell et al., 2021). Second, we excluded

Table 1

Final query used for data collection.

No.	Search Query	Results
#1	Final query: (TITLE (“artificial intelligence” OR “machine learning” OR “machine*learning” OR “learning system*” OR “neural network*” OR “deep learning” OR “support vector machine*” OR “decision tree*” OR “artificial neural network*” OR “supervised learning” OR “random forest” OR “natural language processing” OR “reinforcement learning” OR “text processing” OR “fuzzy logic” OR “Bayesian network*” OR “fuzzy test” OR “stochastic systems” OR “AI” OR “SVR” OR “RFR” OR “LSTM” OR “hft”)) OR KEY (“artificial intelligence” OR “machine learning” OR “machine*learning” OR “learning system*” OR “neural network*” OR “deep learning” OR “support vector machine*” OR “decision tree*” OR “artificial neural network*” OR “supervised learning” OR “random forest” OR “natural language processing” OR “reinforcement learning” OR “text processing” OR “fuzzy logic” OR “Bayesian network*” OR “fuzzy test” OR “stochastic systems” OR “AI” OR “SVR” OR “RFR” OR “LSTM” OR “hft”)) AND (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “re”)) AND (LIMIT-TO (SUBJAREA, “ECON”)) AND (LIMIT-TO (LANGUAGE, “English”)) AND (LIMIT-TO (SRCTYPE, “j”))) Limited online to English, journals, articles and reviews, economics and finance.	5699
#2	Limiting the study to studies published from 2011–2021 and excluding journals that are less ranked than Q1 and Q2, as per the Scopus ranking 2021.	1388
#3	Excluding nonfinancial journals and manual filtration for analysis.	348

nonfinancial journals. Third, we only filtered the journals based on Scopus Q1 and Q2 criteria. Fourth, apart from using manual filtration techniques to exclude irrelevant articles, we conducted an examination of each paper and its relation to the topic.

Overall, the data was prepared for analysis using dedicated bibliometric software. This inclusion and exclusion criteria were contingent on the topic's explicit coverage within the defined domain of the literature. The 348 articles and reviews after the filtration process were used for further bibliometric and content analyses, which we will explain in the next stage. Specifically, these are all of the articles and reviews relevant to AI and ML published in the Scopus Q1 and Q2 finance journals.

Finally, AI (as a science) emerged in the previous century, while ML was coined in 1959 by Arthur Samuel, and grew as a sub-field of AI (Bhavsar et al., 2017). However, the flourish of academic literature in this domain began in the 1990s, while its applications emerged after 2010. Thus, we only chose the time frame based on the noticeable growth of AI and ML applications after 2010.

2.2. Study approach and tools

The bibliometric approach was first introduced by Pritchard (1969) to identify and understand the network of articles based on citations. Since then, bibliometric and scientometric methods have been largely used in previous literature (Goodell et al., 2021). Its superiority stems from the ability to produce accurate and reproducible results, providing readers with full information concerning the intellectual developments of a certain field (Donthu et al., 2021; Linnenluecke et al., 2020). Bibliometrics has also been used to analyze journals (Alshater et al., 2020), subjects and topics (Alshater et al., 2020), countries, universities, and authors' output (Moed et al., 1985). When combined with content analysis, this method summarizes the contributions made to a research topic.

In this study, we first conducted a performance analysis that includes the most relevant scientific actors in the literature. Second, we performed a citation analysis in which we extensively revealed the performance of scientific field actors (i.e., documents, authors, journals, institutions, and countries). Third, we performed a network analysis in which we used VOSviewer and RStudio (biblioshiny) to conduct the keyword co-occurrence, factorial analysis, trend analysis, co-authorships, and bibliographical coupling. In this case, VOSviewer (Van Eck and Waltman, 2010) enriched our explorations of the leading contributors, while RStudio (with the Bibliometrix package) included a graphical interface that allowed us to perform bibliometric tests (Aria and Cuccurullo, 2017). Both programs are widely used to investigate bibliometric data (Alshater et al., 2021; Paltrinieri et al., 2019).

3. Results and discussion

3.1. General information and performance analysis

Our dataset contained 348 documents (336 articles and 12 reviews) from 48 sources, while the time frame of the collected studies was 2011–2021. The average citation per document was 8.61, while the average citation per year was as low as 2.4. Additionally, we detected 839 authors among the 348 documents. The collaboration index in our dataset was 2.68, implying that there were 2.68 authors (on average) in each paper.

Fig. 1 illustrates the distribution of the literature from 2011 to 2021, achieving an average growth rate of 34%. It is noticeable that the research productivity was almost stable from 2011 to 2016. However, considerable growth has occurred since 2017. This trend is expected to grow over the next decade. According to Cioffi et al. (2020), the AI and ML literature has significantly grown since 2013 (Cioffi et al., 2020). However, Fig. 1 shows that the AI and ML literature in the finance field considerably increased in 2018, indicating a lag period of approximately five years, compared to the general literature in AI and ML. This indicates that the finance industry had some time to adopt the newer technology.

Fig. 2 illustrates the most active countries in the AI and ML literature in the finance field. We found that the United States, the United Kingdom, and China were the most productive and impactful countries. This analysis was based on counting each author affiliated with a country. For example, if four authors collaborated in a research article under the U.S. affiliation, then this will credit the United States four times. Thus, we also relied on the corresponding authors for more representation regarding the productivity of

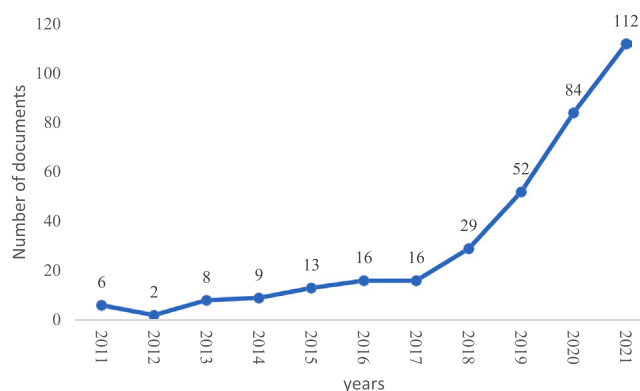


Fig. 1. AI and ML in finance literature growth.

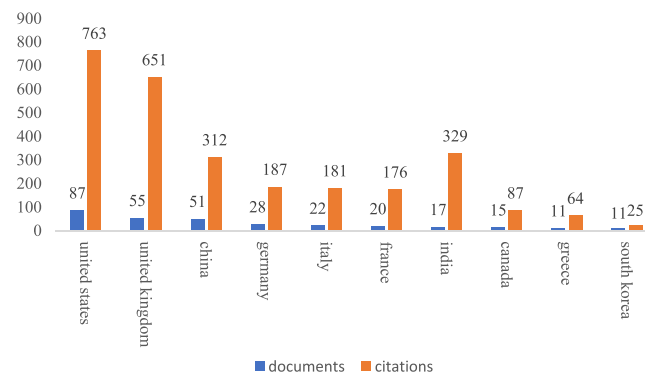


Fig. 2. Most Productive and Impactful Countries.

the countries. Fig. 3 shows the frequency of the most active countries based on the corresponding authors. It also presents the ratios of multiple country publications (MCP) and single country publications (SCP).

Fig. 4 illustrates the most active research institutions in this field. This information is useful for funding bodies, early career researchers, and students searching for postgraduate programs in this area. University College London ranks first on the list, followed by Tishnghua University and the University of Waterloo. Table 2 provides a list of the finance journals that produced the most in the AI and ML literature. This information can be helpful for researchers in identifying prolific sources in this area and finding relative documents based on their interests. Finally, we present the most active researchers in this field. Fig. 5 shows the most relevant authors, while Fig. 6 includes their activeness since 2010.

3.2. Citation analysis

Based on Fig. 7, which presents the citation trend, the average citation per article was eight. However, since the number of articles has been significantly growing since 2017, the citation pattern should show an upward trend over the next five years.

Table 3 presents the top 10 cited articles in order to highlight the most prominent and impactful literature. For example, Khandani et al.'s (2010) study on consumer credit-risk models via ML was cited 192 times, while Chaboud et al. (2014) research on the rise of ML algorithmic trading in the foreign exchange market was cited 180 times. We also found that the top three articles were from the most prestigious journals, including the *Journal of Banking and Finance*, the *Journal of Finance*, and the *Journal of Financial Economics*.

According to Fig. 8, which presents the most impactful sources, we found that the *Journal of Quantitative Finance* was the most impactful, followed by the *Journal of Banking and Finance* and the *Journal of Intelligent Systems in Accounting, Finance, and Management*. In addition, the majority of the cited articles in the literature came from financial rather than technological journals. Table 4 shows the most impactful authors in this area.

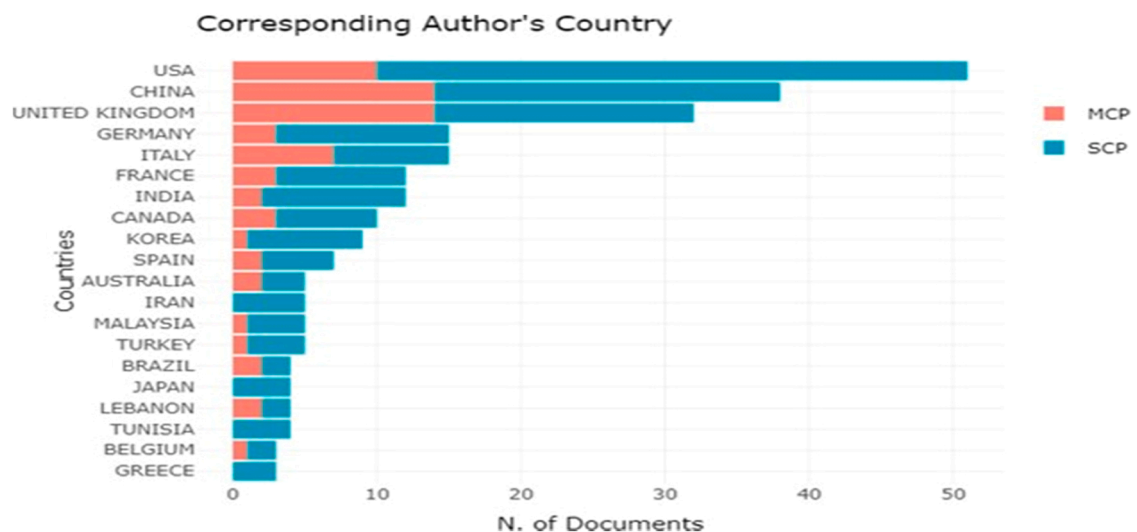


Fig. 3. Country Production Based on the Corresponding Authors. Note: MCP refers to multiple country publications, while SCP refers to the single country publications ratio.

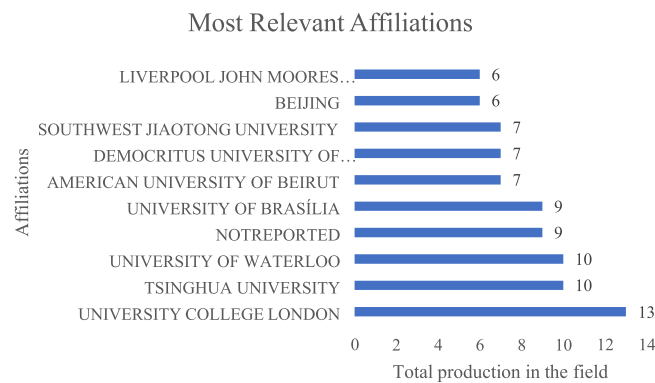


Fig. 4. Most Active Academic institutions.

Table 2
Most relevant academic sources.

#	Sources	Articles
1.	<i>Quantitative Finance</i>	66
2.	<i>Intelligent Systems in Accounting Finance and Management</i>	33
3.	<i>Finance Research Letters</i>	21
4.	<i>Journal of Finance and Data Science</i>	21
5.	<i>Journal of Behavioral and Experimental Finance</i>	19
6.	<i>Journal of Banking and Finance</i>	14
7.	<i>North American Journal of Economics and Finance</i>	13
8.	<i>European Journal of Finance</i>	12
9.	<i>International Journal of Finance and Economics</i>	12
10.	<i>Research In International Business and Finance</i>	11

Note: This table shows the number of publications for each journal based on our criteria of inclusion and exclusion.

3.3. Network visualization

This section presents the co-citation of the sources and the co-authorship of the authors. Fig. 9 presents the co-citation patterns for the sources. The higher (lower) distance between the sources means less (more) connectedness between the sources. We used 81 sources and found that only 39 were connected, which indicates low connectivity between the journals. This is in line with our previous finding of low citations in this field.

Fig. 10 reveals the co-authorship patterns in the literature. We obtained the data from 839 authors and found that only 31 authors were connected. This low co-authorship urges more research groups to work in this field.

3.3.1. Keywords/Cartography analysis

We performed a keyword analysis using VOSviewer software. This test not only reaffirmed our choice of selection but was also an indicator of the clear clusters in the literature. For example, based on Fig. 11, the red cluster connects the main topics and ideas, including AI, ML, deep learning, neural networks, big data, and other relevant technological terminologies, with financial concepts such as implied volatility, portfolio optimization, risk assessment, financial crises, and investment. It is noticeable how these concepts and algorithms were implied in forecasting time series and clustering, which is one of the purposes of AI and ML. Meanwhile, algorithmic trading constitutes a separate clear cluster, with the research in this domain starting much earlier than that in other areas.

Fig. 12 reveals the most recent researchable topics. The new topics are colored in yellow, indicating that the most recent research focused on ML, deep learning, and big data in finance. This map reaffirms the recent rise of AI and ML research in finance.

We also created the three-field plot shown in Fig. 13. In this case, the analysis was limited to 20 sources. The keywords are on the right, the countries are in the middle, and the journals are on the left. Based on this figure, the United States and the United Kingdom dominated this field, followed by China. As for the sources, Quantitative Finance and Risks dominated this area.

3.4. Bibliographic coupling and content analysis

3.4.1. Research Stream 1: bankruptcy prediction and credit-risk assessment

This research stream focuses on applying AI and ML for corporate bankruptcy prediction, credit or default risk assessment, credit scoring, and credit ratings. Bankruptcy prediction has traditionally been a critical exercise of external parties (e.g., investors, creditors, auditors, and regulators) of a firm to predict bankruptcy from accounting-based information (Altman, 1968) and market-based information (Black and Scholes, 2019; Merton, 1974). Thus, the prediction of creditworthiness has been of considerable interest to

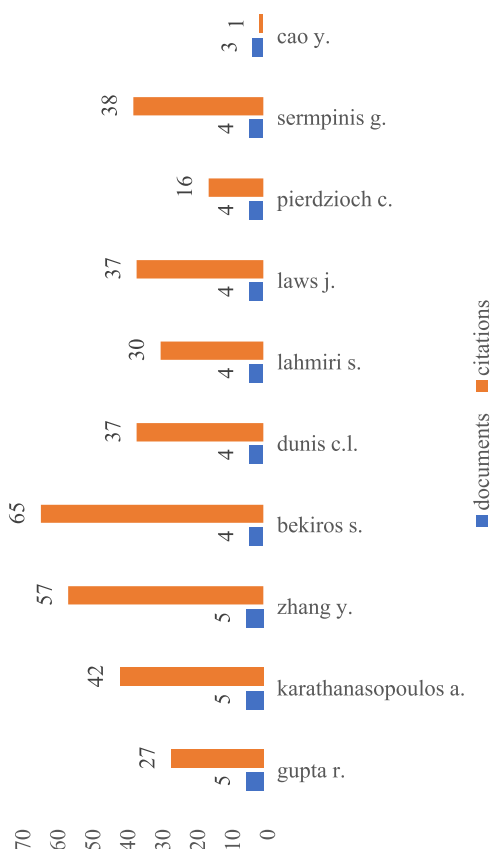


Fig. 5. Most Relevant Authors.

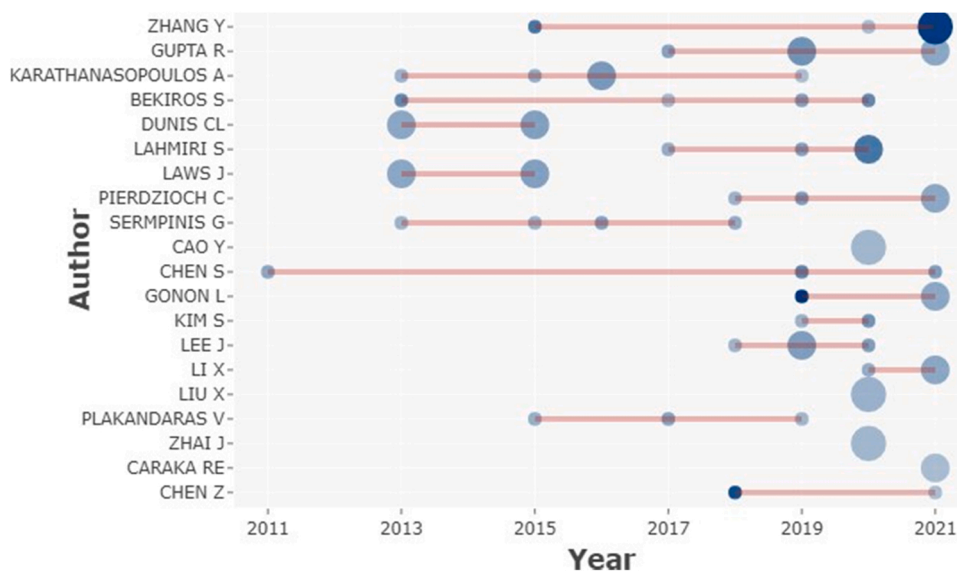


Fig. 6. Authors' Productivity (Output of the Top Authors Over Time). Notes: The larger the circle, more the articles the author published in that year. The darker the circle, more the citations per year.

investors, creditors, regulators, practitioners, and other stakeholders (Altman, 1968, 2002; Jones and Hensher, 2004; Jones and Johnstone, 2012; Jones et al., 2015, 2017).

Credit events in both crisis and non-crisis periods can generate significant economic costs to society (Jones et al., 2017). The world

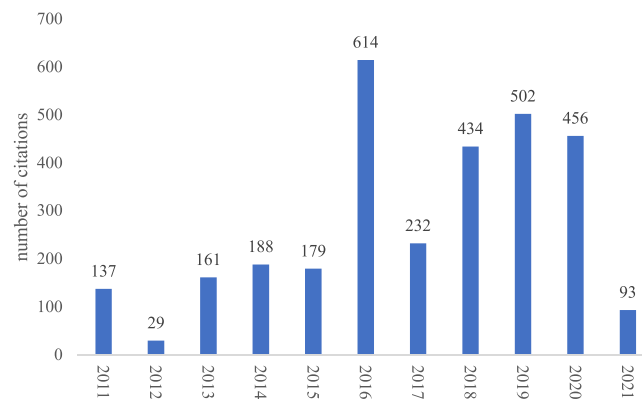


Fig. 7. Citation growth overtime.

Table 3
Most cited documents.

Authors	Title	Year	Source Title	Cited by
Khandani A.E., Kim A.J., Lo A. W.	Consumer credit-risk models via machine- learning algorithms	2010	<i>Journal of Banking and Finance</i>	192
Chaboud A.P., Chiquoine B., Hjalmarsson E., Vega C.	Rise of the machines: Algorithmic trading in the foreign exchange market	2014	<i>Journal of Finance</i>	180
O'Hara M.	High frequency market microstructure	2015	<i>Journal of Financial Economics</i>	167
Moghaddam A.H., Moghaddam M.H., Esfandyari M.	Stock market index prediction using artificial neural network [Predicción del índice del mercado bursátil utilizando una red neuronal artificial]	2016	<i>Journal of Economics, Finance and Administrative Science</i>	163
Fernandes M., Medeiros M.C., Scharth M.	Modeling and predicting the Chicago Board Options Exchange market volatility index	2014	<i>Journal of Banking and Finance</i>	90
Dash R., Dash P.K.	A hybrid stock trading framework integrating technical analysis with machine learning techniques	2016	<i>Journal of Finance and Data Science</i>	82
Henrique B.M., Sobreiro V.A., Kimura H.	Stock price prediction using support vector regression on daily and up to the minute prices	2018	<i>Journal of Finance and Data Science</i>	78
Zhong G., Wang L.-N., Ling X., Dong J.	An overview on data representation learning: From traditional feature learning to recent deep learning	2016	<i>Journal of Finance and Data Science</i>	77
Bekhet H.A., Eletter S.F.K.	Credit-risk assessment model for Jordanian commercial banks: Neural scoring approach	2014	<i>Review of Development Finance</i>	65
Basak S., Kar S., Saha S., Khaidem L., Dey S.R.	Predicting the direction of stock market prices using tree-based classifiers	2019	<i>North American Journal of Economics and Finance</i>	62

has witnessed a considerable number of credit events in recent decades, in particular, the Asian financial crisis in 1997–1998, the dot.com bubble in 2000–2002, the global financial crisis in 2007–2009, and the European debt crisis in 2010–2012. In this regard, there has been a significant number of studies on the prediction of creditworthiness (Aziz and Dar, 2006; Flagg et al., 1991; Johnsen and Melicher, 1994; Li and Faff, 2019).

While extensive literature on creditworthiness prediction has emerged, the methodologies have included simple classification models, such as linear discriminant analysis (LDA), binary logistic or probit models, and multinomial logit models (see Altman, 1968; Altman et al., 1977; Jones and Hensher, 2004; Ohlson, 1980), and “new age” classifications models, such as AdaBoost, which have generalized boosting and random forests (see Jones et al., 2017). The prediction of creditworthiness with “new age” models, as seen in Jones et al. (2017), has significantly outperformed the simple classification models. They have included LDA, logistic or probit models, and advanced data mining techniques such as neural networks and support vector machines (SVMs) (see Friedman, 2001; Jones et al., 2015).

Hernandez Tinoco and Wilson (2013) used accounting, market-based, and macroeconomic data to predict corporate bankruptcy, benchmarking the results of the panel logit model against the application of artificial neural networks (ANNs) and Altman's (1968) model. Kwak et al. (2012) applied the multiple criteria linear program (MCLP) as well as ML approaches, such as decision trees and SVMs, to predict the bankruptcy of Korean firms. The results from both the MCLP and ML approaches showed a similar overall prediction accuracy of bankruptcy. For example, Lahmiri and Bekiros (2019) found that the application of nonlinear classifiers based on big data analytics and ML approaches outperformed the prediction results, compared to the traditional approaches of bankruptcy prediction, while Chen et al. (2011) applied SVM to predict the bankruptcy of German firms.

In related studies, Khandani et al. (2010) used ML techniques to assess consumer credit risk for a sample of major commercial banks and found that the application of ML techniques significantly improved the prediction of credit card holder delinquencies and defaults. Trustorff et al. (2011) applied SVMs to estimate default classification and default probability, while Acheampong and Elshandidy

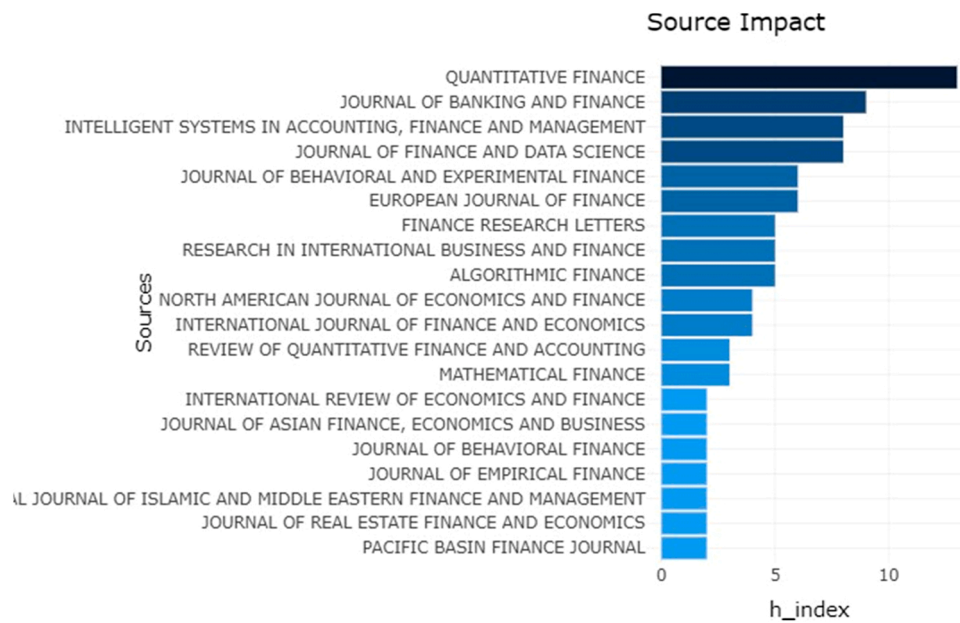


Fig. 8. Impactful academic sources.

Table 4
Authors' Impact.

Author	h-index	g-index	m-index	TC	NP	PY_start
Karathanasopoulos	4	5	0.444	42	5	2013
Dunis	4	4	0.444	37	4	2013
Laws	4	4	0.444	37	4	2013
Sermpinis	4	4	0.444	38	4	2013
Zhang	3	5	0.429	57	5	2015
Bekiros	3	4	0.333	65	4	2013
Lahmiri	3	4	0.6	30	4	2017
Chen	3	3	0.273	43	3	2011
Gupta	2	5	0.4	27	5	2017
Pierdzioch	2	4	0.5	16	4	2018

Notes: This table reports the top authors. The g-index indicates that the top 10 publications of an author have been cited at least 100 times. The m-index is the h-index divided by the number of years that a researcher has been active. TC = total citations, NP = Number of publications, and PY_Start = the starting year of publication.

(2021) used a ML algorithm to extract relevant information from the annual reports of borrowing firms and examined whether such information helped predict borrowers' default risk.

Furthermore, De Moor et al. (2018) used ML techniques to evaluate the subjective component of sovereign credit ratings. They found that subjective judgment has a predictive value in determining such ratings. Plakandaras et al. (2020) applied SVMs to forecast the credit ratings of European banking institutions, while Kočenda and Vojtek (2011) used both logistic regression models and ML approaches to estimate the retail credit scores from loan-level data. Abdou et al. (2014) applied the neural network approach to evaluate the credit scoring model of the Islamic financial houses in United Kingdom, while Huang et al. (2004) applied SVMs and neural networks to predict the bond ratings in the European debt market. Finally, Kim and Han (2001) used ANNs for the prediction of corporate bond ratings.

3.4.2. Research Stream 2: stock price prediction, portfolio management, volatility, and liquidity

Predicting stock prices is critical for investors and it has been one of the most popular research topics. Although traditional models have been dominant in predicting stock returns (Barndorff-Nielsen et al., 2009; Hansen et al., 2012), ML techniques have been popular in predicting stock prices over the last decade (Ghosh et al., 2021). For instance, Gorenc Novak and Velušček (2016) applied ML approaches, such as SVMs, to predict stock prices and found that ML techniques significantly increase prediction accuracy, compared to traditional models. Similarly, Ghosh et al. (2021) employed both random forests and long short-term memory (LSTM) techniques to predict the prices of S&P 500 firms, while Liu et al. (2021) applied the LSTM model to predict the Shanghai Stock Exchange (SSE) 50 firms.

Equally, ML approaches have been popular for constructing portfolios, predicting stock volatility, and measuring stock liquidity.

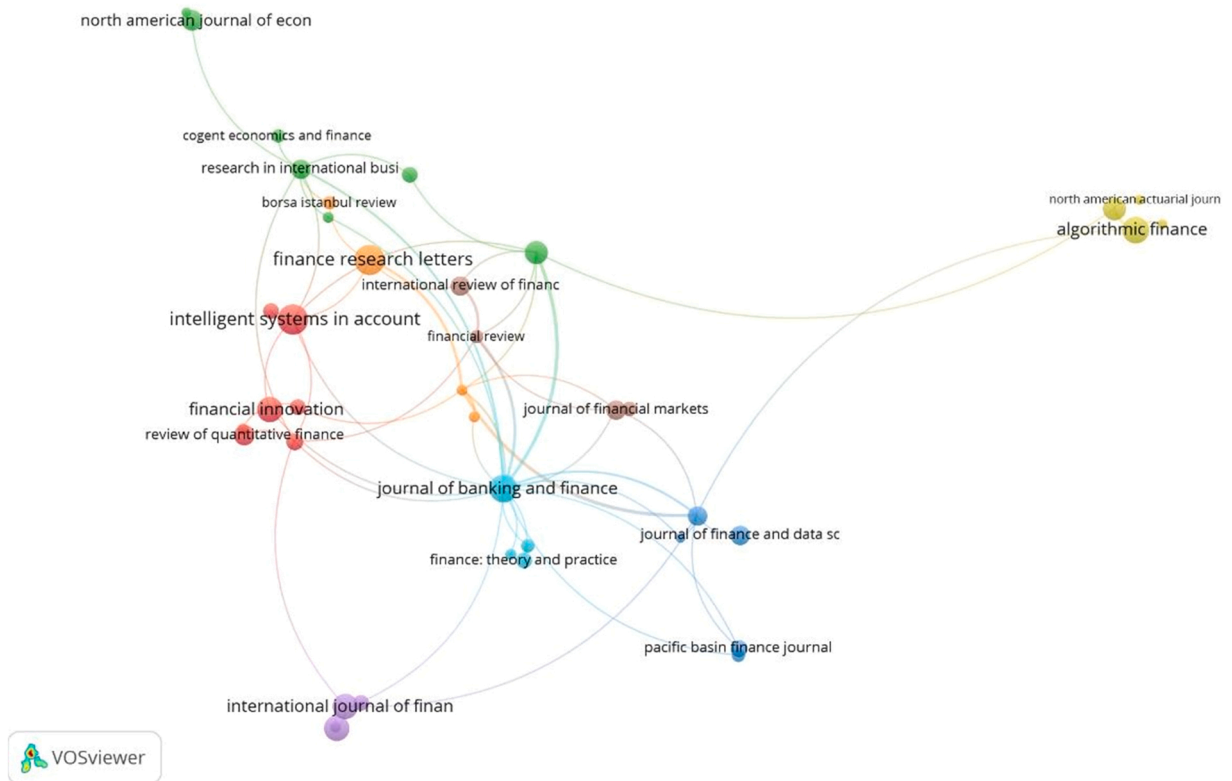


Fig. 9. Co-citations of Academic Sources. Notes: This figure has been created using VOSviewer. The network map shows the co-citation patterns of the 29 (out of the 81) journals. The criteria included mapping the journals with at least 10 citations from the studies that we reviewed. The node size corresponds to the number of citations, while the lines correspond to the existence of a citation in either direction. The distance between the nodes corresponds to the tendency for journals to be cited by other studies.

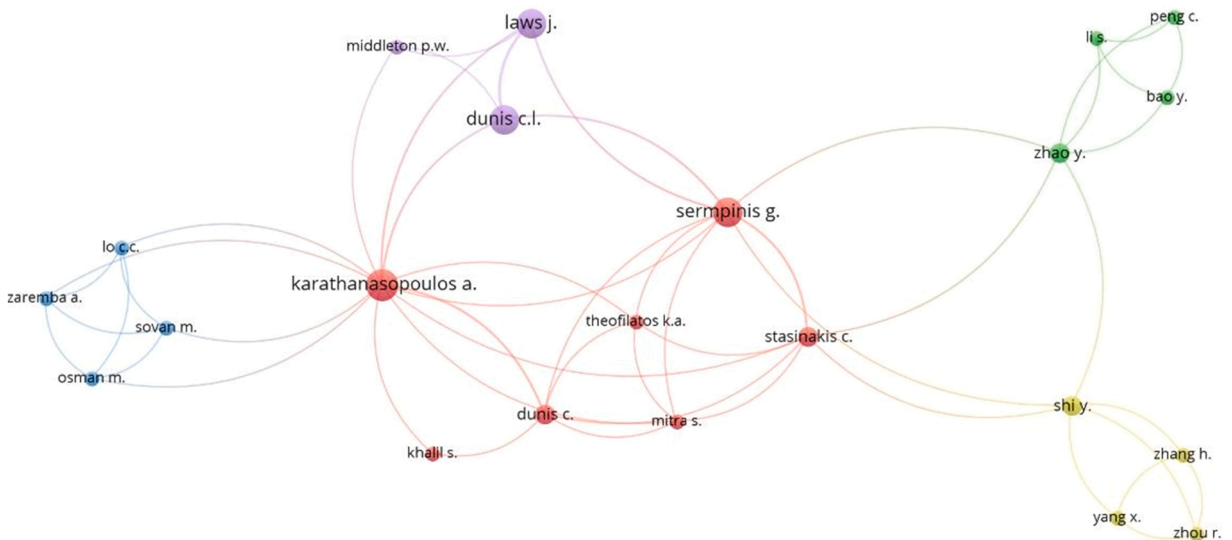


Fig. 10. Co-authorship for authors. Notes: Co-authorship refers to those who work in tandem to write a piece of scientific research. This figure describes the patterns of co-authorship among the top cited researchers.

For example, [Manela and Moreira \(2017\)](#) used support vector regression (SVR) to construct a text-based measure of stock volatility using the front-page titles of the *Wall Street Journal*. Moreover, [Chen and Ge \(2021\)](#) applied the learning-based strategy for optimal investments using neural networks, while [Braun et al. \(2020\)](#) used ANNs to analyze stock liquidity.

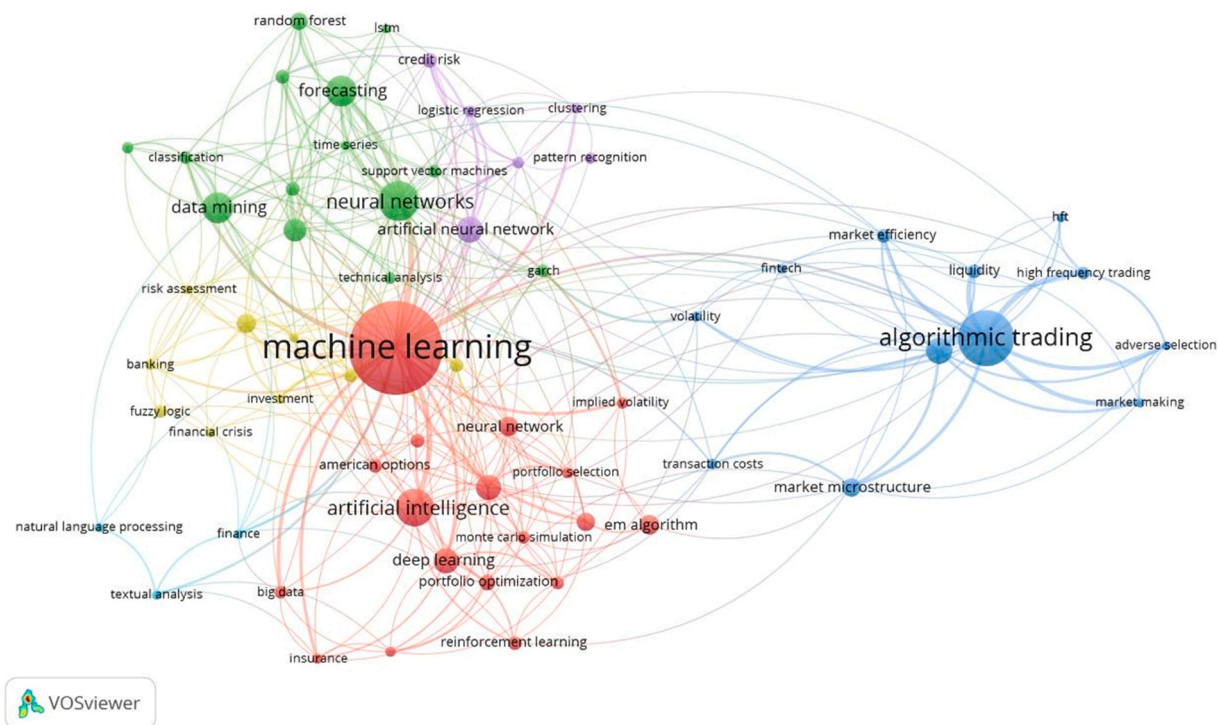


Fig. 11. Keyword co-occurrence. Notes: The size of the nodes indicates the frequency of occurrence, while the curves between the nodes represent their co-occurrence. The shorter the distance between the two nodes, the larger the number of co-occurrences of the two keywords. This figure can help identify the content clusters.

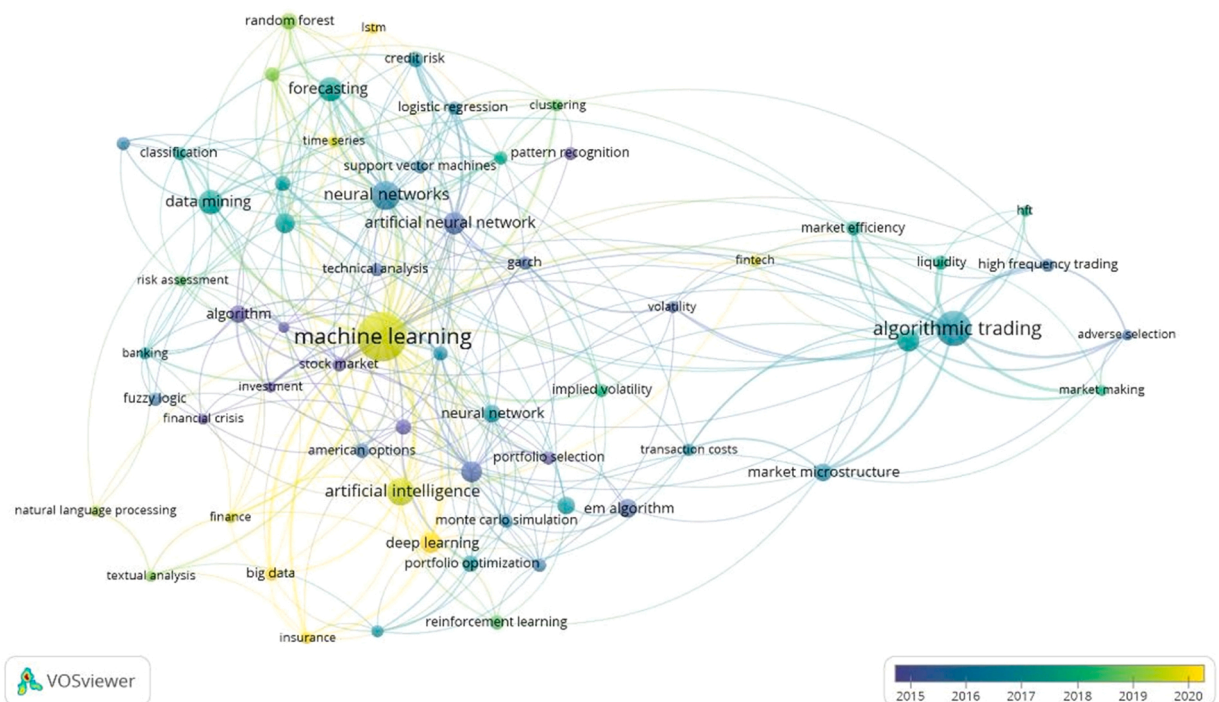


Fig. 12. Keyword overlay visualization. Notes: The size of the nodes indicates the frequency of occurrence, while the curves between the nodes represent their co-occurrence. The shorter the distance between the two nodes, larger the number of co-occurrences of the two keywords. The more yellow the color, the trendier the topics in the literature.

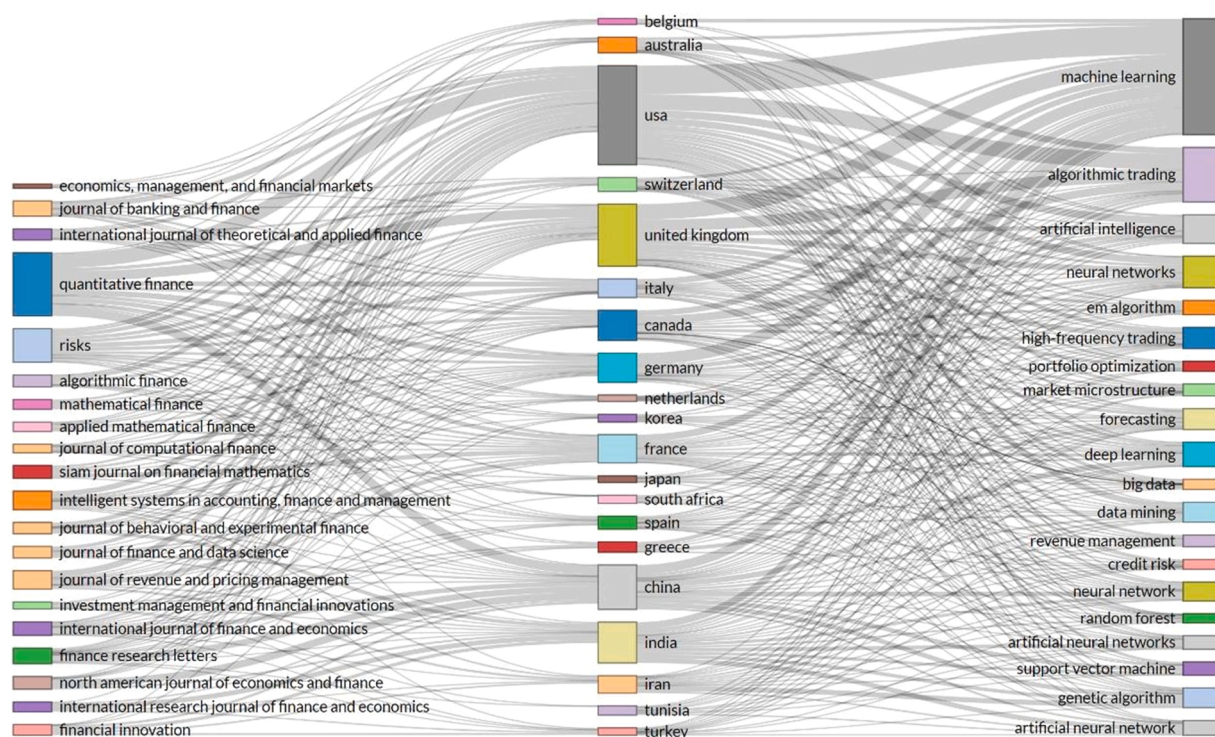


Fig. 13. Three-field plot. Notes: This presents the connectedness between the journals, countries, and keywords. It was created to depict the proportion of the participation of each country, with specific journals and topics represented by the keywords.

3.4.3. Research Stream 3: prediction of the prices of oil, gold, and agriculture products

Changes in oil prices can affect economic activities and consumer sentiments (Hamilton, 2009). Hence, predicting such changes is critical for the economy. In this regard, ANNs have been used to capture the nonlinearities in oil price dynamics (Dbouk and Jamali, 2018). For example, Verma (2021) combined the generalized autoregressive conditional heteroscedasticity (GARCH) and Glosten-Jagannathan-Runkle (GJR)-GARCH models with the LSTM model to predict crude oil volatility and examine the performances of the hybrid model, compared to those of the classical volatility prediction models.

ML techniques have also been frequently applied to predict the prices of gold and agriculture products. For instance, Malliaris and Malliaris (2013) used a neural network model to examine the relationship between gold, oil, and the euro and found that oil and its lagged values can be used as predictors. Furthermore, Manogna and Mishra (2021) applied deep learning techniques, such as the time-delay neural network (TDNN) and LSTM, to forecast the prices of agricultural commodities, and found that LSTM can predict prices, compared to TDNN and other linear models. In a related study, Ghahari et al. (2019) used deep learning algorithms, such as deep belief networks, and found higher prediction accuracy.

3.4.4. Research Stream 4: anti-money laundering, anti-fraud detection, and risk management

Money laundering has become an issue of international concern since the 9/11 event (Unger, 2009). Specifically, detecting money laundering has been a major objective of governments across the world in order to combat terrorism financing. Previous research has shown that ML techniques have been used to detect money laundering activities (Garcia-Bedoya et al., 2020; Jullum et al., 2020; Kannan and Somasundaram, 2017). Jullum et al. (2020) developed and validated a ML model to detect such activities in Norway's largest bank (the DNB), while Singh and Lin (2020) suggested that ML techniques can help detect money laundering activities in the context of charity organizations in the United Kingdom.

Finally, ML techniques have become popular tools for detecting frauds and managing risks (Eling et al., 2021; Omar et al., 2017). In related studies, Omar et al. (2017) applied an ANN to predict fraudulent financial reporting and found that the model outperformed traditional statistical techniques, while Goel and Uzuner (2016) used SVMs to construct a fraud sentiment model from the Management Discussion and Analysis (MD & A) sections of annual reports.

3.4.5. Research Stream 5: behavioral finance

AI has been widely used to model behavioral biases (Bhatia et al., 2020; Königstorfer and Thalmann, 2020a, 2020b). Specifically, Königstorfer and Thalmann (2020a, 2020b) found that commercial banks can lower credit risk, automate compliance tasks, enhance security in processing payments, and improve customer retention by using AI. Moreover, Ghosh and Kozarevic (2018) applied ML techniques to identify possible explosive moments in the CNX NIFTY Index (a major stock index in India), while Félix et al. (2020) used

ML-based models, such as random forests and deep neural networks, to construct implied volatility sentiment.

3.4.6. Research Stream 6: big data analytics, blockchain, and data mining

The PwC Global Fintech Report 2019 showed that AI, blockchain, the Internet of Things, and big data analytics are among the top 10 emerging technologies for the financial services industry. In this regard, [Hanafy and Ming \(2021\)](#) applied ML techniques, such as random forest, decision trees, and XGBoost, to improve customer service, while [Antoncic \(2020\)](#) demonstrated that corporations can use AI and big data analytics in their business strategies in order to gain a competitive advantage over their peers. Furthermore, [Huang et al. \(2017\)](#) used ML techniques, such as decision trees, neural networks, SVMs, and a generalized multiple kernel machine (as a data mining tool) to predict bankruptcy.

4. Conclusion

This study reviewed the AI and ML literature in the finance field. Using a bibliometric approach, we analyzed 348 articles published in 2011–2021 from the Q1 and Q2 finance journals indexed in the Scopus database. Our analysis showed an upward trajectory in the publication trend starting from 2015. Additionally, the popular research topics on how AI has been adopted included bankruptcy prediction, stock price prediction, portfolio management, oil price prediction, anti-money laundering, behavioral finance, big data analytics, and blockchain. As for the top three countries contributing to the literature, they were the United States, the United Kingdom, and China. It should be noted that since our study did not provide a performance comparison of different ML techniques and deep ensembles, future research should emphasize and compare the performance of different ML techniques and deep ensembles in the field of finance.

CRedit authorship contribution statement

Shamima Ahmed: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Muneer M. Alshater:** Conceptualization, Methodology, Formal analysis, Data curation, Formal Investigation. **Anis El Ammari:** Resources, Writing – review & editing. **Helmi Hammami:** Formal analysis, Resources.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ribaf.2022.101646](https://doi.org/10.1016/j.ribaf.2022.101646).

References

- Abdou, A.H., Alam, S.T., Mulkeen, J., 2014. Would credit scoring work for Islamic finance? A neural network approach. *Int. J. Islam. Middle East. Financ. Manag.* 7 (1), 112–125. <https://doi.org/10.1108/IMEFM-03-2013-0038>.
- Acheampong, A., Elshandidy, T., 2021. Does soft information determine credit risk? Text- based evidence from European banks. *J. Int. Financ. Mark. Inst. Money* Art., 101303 <https://doi.org/10.1016/j.intfin.2021.101303>.
- Alshater, M.M., Atayah, O.F., Hamdan, A., 2021. Journal of sustainable finance and investment: a bibliometric analysis. *J. Sustain. Financ. Invest.* 1–22.
- Alshater, M.M., Hassan, M.K., Khan, A., Saba, I., 2020. Influential and intellectual structure of Islamic finance: A bibliometric review. *Int. J. Islam. Middle East. Financ. Manag.*
- Altman, E.I., 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *J. Financ.* 23 (4), 589–609.
- Altman, E.I., 2002. *Bankruptcy, Credit Risk, and High Yield Junk Bonds*. Wiley-Blackwell.
- Altman, E.I., Haldeman, R.G., Narayanan, P., 1977. A new model to identify bankruptcy risk of corporations. *J. Bank. Financ.* 1 (1), 29–54.
- Antoncic, M., 2020. A paradigm shift in the board room: Incorporating sustainability into corporate governance and strategic decision-making using big data and artificial intelligence. *J. Risk Manag. Financ. Inst.* 13 (4), 290–294. (<https://www.scopus.com/inward/record.uri?eid=2-s2.0-85097233644&partnerID=40&md5=9e358ebf5745e6363b5de08bf42ab63e>).
- Aria, M., Cuccurullo, C., 2017. Bibliometrix: An R-tool for comprehensive science mapping analysis. *J. Informetr.* 11 (4), 959–975.
- Aziz, M.A., Dar, H.A., 2006. Predicting corporate bankruptcy: Where we stand? *Corp. Gov.: Int. J. Bus. Soc.*
- Bahrammirzaee, A., 2010. A comparative survey of artificial intelligence applications in finance: Artificial neural networks, expert system and hybrid intelligent systems. *Neural Comput. Appl.* 19 (8), 1165–1195.
- Baker, H.K., Kumar, S., Pattnaik, D., 2020. Fifty years of the financial review: a bibliometric overview. *Financ. Rev.* 55 (1), 7–24.
- Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N., 2009. Realized kernels in practice: Trades and quotes. *Econ. J.* 119 (3), C1–C32. <https://doi.org/10.1111/j.1368-423X.2008.00275.x>.
- Bhatia, A., Chandani, A., Chhateja, J., 2020. Robo advisory and its potential in addressing the behavioral biases of investors—A qualitative study in Indian context. *J. Behav. Exp. Financ.* 25 <https://doi.org/10.1016/j.jbef.2020.100281>.
- Bhavsar, P., Safo, I., Bouaynaya, N., Polikar, R., Dera, D., 2017. Machine learning in transportation data analytics. *Data analytics for intelligent transportation systems*. Elsevier, pp. 283–307.
- Black, F., & Scholes, M. (2019). The pricing of options and corporate liabilities. In *World Scientific Reference on Contingent Claims Analysis in Corporate Finance: Volume 1: Foundations of CCA and Equity Valuation* (pp. 3–21). World Scientific.
- Braun, J., Hausler, J., Schäfers, W., 2020. Artificial intelligence, news sentiment, and property market liquidity. *J. Prop. Invest. Financ.* 38 (4), 309–325. <https://doi.org/10.1108/JPIF-08-2019-0100>.
- Bughin, J., Seong, J., Manyika, J., Chui, M., Joshi, R., 2018. Notes from the AI frontier: Modeling the Impact of AI on the World Economy. McKinsey Global Institute., Chaboud, A.P., Chiquoine, B., Hjalmarsson, E., Vega, C., 2014. Rise of the machines: Algorithmic trading in the foreign exchange market. *J. Financ.* 69 (5), 2045–2084. <https://doi.org/10.1111/jofi.12186>.
- Chen, S., Ge, L., 2021. A learning-based strategy for portfolio selection. *Int. Rev. Econ. Financ.* 71, 936–942. <https://doi.org/10.1016/j.iref.2020.07.010>.

- Chen, S., Härdle, W.K., Moro, R.A., 2011. Modeling default risk with support vector machines. *Quant. Financ.* 11 (1), 135–154. <https://doi.org/10.1080/14697680903410015>.
- Cioffi, R., Travaglioni, M., Piscitelli, G., Petrillo, A., De Felice, F., 2020. Artificial intelligence and machine learning applications in smart production: Progress, trends, and directions. *Sustain. (Switz.)* 12 (2), 492.
- Dbouk, W., Jamali, I., 2018. Predicting daily oil prices: Linear and non-linear models. *Res. Int. Bus. Financ.* 46, 149–165. <https://doi.org/10.1016/j.ribaf.2018.01.003>.
- De Moor, L., Luitel, P., Sercu, P., Vanpée, R., 2018. Subjectivity in sovereign credit ratings. *J. Bank. Financ.* 88, 366–392. <https://doi.org/10.1016/j.jbankfin.2017.12.014>.
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., Lim, W.M., 2021. How to conduct a bibliometric analysis: An overview and guidelines. *J. Bus. Res.* 133, 285–296.
- Eling, M., Nuessle, D., Staubli, J., 2021. The impact of artificial intelligence along the insurance value chain and on the insurability of risks. *Geneva Pap. Risk Insur.: Issues Pract.* <https://doi.org/10.1057/s41288-020-00201-7>.
- Félix, L., Kräussl, R., Stork, P., 2020. Implied volatility sentiment: A tale of two tails. *Quant. Financ.* 20 (5), 823–849. <https://doi.org/10.1080/14697688.2019.1696018>.
- Flagg, J.C., Giroux, G.A., Wiggins Jr., C.E., 1991. Predicting corporate bankruptcy using failing firms. *Rev. Financ. Econ.* 1 (1), 67–75.
- Friedman, J.H., 2001. Greedy function approximation: A gradient boosting machine. *Ann. Stat.* 1189–1232.
- Garcia-Bedoya, O., Granados, O., Cardozo Burgos, J., 2020. AI against money laundering networks: The Colombian case. *J. Money Laund. Control* 24 (1), 49–62. <https://doi.org/10.1108/JMLC-04-2020-0033>.
- Ghahari, A., Newlands, N.K., Lyubchich, V., Gel, Y.R., 2019. Deep learning at the interface of agricultural insurance risk and spatio-temporal uncertainty in weather extremes. *North Am. Actuar. J.* 23 (4), 535–550. <https://doi.org/10.1080/10920277.2019.1633928>.
- Ghosh, B., Kozarevic, E., 2018. Identifying explosive behavioral trace in the CNX nifty index: A quantum finance approach. *Invest. Manag. Financ. Innov.* 15 (1), 208–223. [https://doi.org/10.21511/imfi.15\(1\).2018.18](https://doi.org/10.21511/imfi.15(1).2018.18).
- Ghosh, P., Neufeld, A., Sahoo, J.K., 2021. Forecasting directional movements of stock prices for intraday trading using LSTM and random forests. *Financ. Res. Lett.* 102280 <https://doi.org/10.1016/j.frl.2021.102280>.
- Goel, S., Uzuner, O., 2016. Do sentiments matter in fraud detection? Estimating semantic orientation of annual reports. *Intell. Syst. Account., Financ. Manag.* 23 (3), 215–239. <https://doi.org/10.1002/isaf.1392>.
- Goodell, J.W., Kumar, S., Lim, W.M., Pattnaik, D., 2021. Artificial intelligence and machine learning in finance: Identifying foundations, themes, and research clusters from bibliometric analysis. *J. Behav. Exp. Financ. Res.* 32, 100577 <https://doi.org/10.1016/j.jbef.2021.100577>.
- Gorenc Novak, M., Velušček, D., 2016. Prediction of stock price movement based on daily high prices. *Quant. Financ.* 16 (5), 793–826. <https://doi.org/10.1080/14697688.2015.1070960>.
- Gupta, A., Dengre, V., Kheruwalla, H.A., Shah, M., 2020. Comprehensive review of text-mining applications in finance. *Financ. Innov.* 6 (1), 1–25.
- Hamilton, J.D., 2009. Causes and Consequences of the Oil Shock of 2007–08. National Bureau of Economic Research.
- Hanafy, M., Ming, R., 2021. Machine learning approaches for auto insurance big data. *Risks* 9 (2), 1–23. <https://doi.org/10.3390/risks9020042>.
- Hansen, P.R., Huang, Z., Shek, H.H., 2012. Realized GARCH: A joint model for returns and realized measures of volatility. *J. Appl. Econ.* 27 (6), 877–906.
- Hernandez Tinoco, M., Wilson, N., 2013. Financial distress and bankruptcy prediction among listed companies using accounting, market and macroeconomic variables. *Int. Rev. Financ. Anal.* 30, 394–419. <https://doi.org/10.1016/j.irfa.2013.02.013>.
- Hilpisch, Y., 2020. Artificial Intelligence in Finance. O'Reilly Media.
- Huang, S.C., Wu, T.K., Wang, N.Y., 2017. An intelligent system for business data mining. *Glob. Bus. Financ. Rev.* 22 (2), 1–7. <https://doi.org/10.17549/gbfr.2017.22.2.1>.
- Huang, Z., Chen, H., Hsu, C.-J., Chen, W.-H., Wu, S., 2004. Credit rating analysis with support vector machines and neural networks: A market comparative study. *Decis. Support Syst.* 37 (4), 543–558. [https://doi.org/10.1016/S0167-9236\(03\)00086-1](https://doi.org/10.1016/S0167-9236(03)00086-1).
- Johnsen, T., Melicher, R.W., 1994. Predicting corporate bankruptcy and financial distress: Information value added by multinomial logit models. *J. Econ. Bus.* 46 (4), 269–286.
- Jones, S., Hensher, D.A., 2004. Predicting firm financial distress: A mixed logit model. *Account. Rev.* 79 (4), 1011–1038.
- Jones, S., Johnstone, D., 2012. Analyst recommendations, earnings forecasts and corporate bankruptcy: Recent evidence. *J. Behav. Financ.* 13 (4), 281–298.
- Jones, S., Johnstone, D., Wilson, R., 2015. An empirical evaluation of the performance of binary classifiers in the prediction of credit ratings changes. *J. Bank. Financ.* 56, 72–85.
- Jones, S., Johnstone, D., Wilson, R., 2017. Predicting corporate bankruptcy: An evaluation of alternative statistical frameworks. *J. Bus. Financ. Account.* 44 (1–2), 3–34.
- Jullum, M., Løland, A., Huseby, R.B., Ånonsen, G., Lorentzen, J., 2020. Detecting money laundering transactions with machine learning. *J. Money Laund. Control* 23 (1), 173–186. <https://doi.org/10.1108/JMLC-07-2019-0055>.
- Kannan, S., Somasundaram, K., 2017. Autoregressive-based outlier algorithm to detect money laundering activities. *J. Money Laund. Control* 20 (2), 190–202. <https://doi.org/10.1108/JMLC-07-2016-0031>.
- Khandani, A.E., Kim, A.J., Lo, A.W., 2010. Consumer credit-risk models via machine-learning algorithms. *J. Bank. Financ.* 34 (11), 2767–2787. <https://doi.org/10.1016/j.jbankfin.2010.06.001>.
- Kim, K.-S., Han, I., 2001. The cluster-indexing method for case-based reasoning using self-organizing maps and learning vector quantization for bond rating cases. *Expert Syst. Appl.* 21 (3), 147–156.
- Kočenda, E., Vojtek, M., 2011. Default predictors in retail credit scoring: Evidence from Czech banking data. *Emerg. Mark. Financ. Trade* 47 (6), 80–98. <https://doi.org/10.2753/REE1540-496X470605>.
- Königstorfer, F., Thalmann, S., 2020a. Applications of artificial intelligence in commercial banks—A research agenda for behavioral finance, 2020/09/01/. *J. Behav. Exp. Financ.* 27, 100352. <https://doi.org/10.1016/j.jbef.2020.100352>.
- Königstorfer, F., Thalmann, S., 2020b. Applications of artificial intelligence in commercial banks—A research agenda for behavioral finance. *J. Behav. Exp. Financ.* 27 <https://doi.org/10.1016/j.jbef.2020.100352>.
- Kwak, W., Shi, Y., Kou, G., 2012. Bankruptcy prediction for Korean firms after the 1997 financial crisis: Using a multiple criteria linear programming data mining approach. *Rev. Quant. Financ. Account.* 38 (4), 441–453. <https://doi.org/10.1007/s11156-011-0238-z>.
- Lahmiri, S., Bekiros, S., 2019. Can machine learning approaches predict corporate bankruptcy? Evidence from a qualitative experimental design. *Quant. Financ.* 19 (9), 1569–1577. <https://doi.org/10.1080/14697688.2019.1588468>.
- Li, L., Faff, R., 2019. Predicting corporate bankruptcy: What matters? *Int. Rev. Econ. Financ.* 62, 1–19.
- Linnenluecke, M.K., Marrone, M., Singh, A.K., 2020. Conducting systematic literature reviews and bibliometric analyses. *Aust. J. Manag.* 45 (2), 175–194.
- Liu, K., Zhou, J., Dong, D., 2021. Improving stock price prediction using the long short-term memory model combined with online social networks. *Article 100507 J. Behav. Exp. Financ.* 30. <https://doi.org/10.1016/j.jbef.2021.100507>.
- Malliaris, A.G., Malliaris, M., 2013. Are oil, gold and the euro inter-related? Time series and neural network analysis. *Rev. Quant. Financ. Account.* 40 (1), 1–14. <https://doi.org/10.1007/s11156-011-0265-9>.
- Mamoshina, P., Ojomoko, L., Yanovich, Y., Ostrovski, A., Botezatu, A., Prikhodko, P., Izumchenko, E., Aliper, A., Romantsov, K., Zhebrak, A., 2018. Converging blockchain and next-generation artificial intelligence technologies to decentralize and accelerate biomedical research and healthcare. *Oncotarget* 9 (5), 5665.
- Manela, A., Moreira, A., 2017. News implied volatility and disaster concerns. *J. Financ. Econ.* 123 (1), 137–162. <https://doi.org/10.1016/j.jfineco.2016.01.032>.
- Manogna, R.L., Mishra, A.K., 2021. Forecasting spot prices of agricultural commodities in India: Application of deep-learning models [Article]. *Intell. Syst. Account., Financ. Manag.* 28 (1), 72–83. <https://doi.org/10.1002/isaf.1487>.
- Merton, R.C., 1974. On the pricing of corporate debt: The risk structure of interest rates. *J. Financ.* 29 (2), 449–470.
- Mhlanga, D., 2020. Industry 4.0 in finance: The impact of artificial intelligence (ai) on digital financial inclusion. *Int. J. Financ. Stud.* 8 (3), 45.

- Moed, H.F., Burger, W., Frankfort, J., Van Raan, A.F., 1985. The use of bibliometric data for the measurement of university research performance. *Res. Policy* 14 (3), 131–149.
- Ohlson, J.A., 1980. Financial ratios and the probabilistic prediction of bankruptcy. *J. Account. Res.* 109–131.
- Omar, N., Johari, Z.A., Smith, M., 2017. Predicting fraudulent financial reporting using artificial neural network. *J. Financ. Crime.* 24 (2), 362–387. <https://doi.org/10.1108/JFC-11-2015-0061>.
- Paltrinieri, A., Hassan, M.K., Bahoo, S., Khan, A., 2019. A bibliometric review of sukuk literature. *Int. Rev. Econ. Financ.*
- Plakandaras, V., Gogas, P., Papadimitriou, T., Doumpa, E., Stefanidou, M., 2020. Forecasting credit ratings of EU banks. *Int. J. Financ. Stud.* 8 (3), 49. <https://doi.org/10.3390/ijfs8030049>.
- Pritchard, A., 1969. Statistical bibliography or bibliometrics. *J. Doc.* 25 (4), 348–349.
- Singh, C., Lin, W., 2020. Can artificial intelligence, RegTech and CharityTech provide effective solutions for anti-money laundering and counter-terror financing initiatives in charitable fundraising. *J. Money Laund. Control.* <https://doi.org/10.1108/JMLC-09-2020-0100>.
- Trustorff, J.H., Konrad, P.M., Leker, J., 2011. Credit risk prediction using support vector machines. *Rev. Quant. Financ. Account.* 36 (4), 565–581. <https://doi.org/10.1007/s11156-010-0190-3>.
- Unger, B., 2009. Money laundering- A newly emerging topic on the international agenda. *Rev. Law Econ.* 5 (2), 807–819. <https://doi.org/10.2202/1555-5879.1417>.
- Van Eck, N.J., Waltman, L., 2010. Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics* 84 (2), 523–538.
- Verma, S., 2021. Forecasting volatility of crude oil futures using a GARCH–RNN hybrid approach. *Intell. Syst. Account., Financ. Manag.* 28 (2), 130–142. <https://doi.org/10.1002/isaf.1489>.
- Wong, B.K., Selvi, Y., 1998. Neural network applications in finance: A review and analysis of literature (1990–1996), 1998/10/01/. *Inf. Manag.* 34 (3), 129–139. [https://doi.org/10.1016/S0378-7206\(98\)00050-0](https://doi.org/10.1016/S0378-7206(98)00050-0).